Theoretical Optics — Rules, Theories & Solving Tips

WAA · Board Review 2026 · Personal study notes

The high-yield ideas behind every question in the reviewer. Pair this with `theoretical-optics-reviewer.html` for drill practice.

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1 · Light, Speed & Index of Refraction

The one equation everything hangs on

`n = c / v` so `v = c / n`

n is the index of refraction (how much a material slows light), c is the speed of light in vacuum, v is the speed inside the material. A bigger n means light crawls slower through that material.

> **Memory hook:** think of n as how "thick" the medium is to light. Diamond (n=2.42) is the thickest of the common ones, so light is slowest there. Vacuum and air are the thinnest, so light runs fastest.

Memorize the speed of light in three units

`c = 186,000 mi/s = 3 x 10⁸ m/s = 300,000 km/s`

Half the speed problems just ask you to divide this by an index. Speed in diamond = 186,000 / 2.42 = 76,955 mi/s. Index from a given speed = 186,000 / speed.

> **Unit trap:** the wrong answer is usually the right number with the wrong unit (km instead of mi, or per minute instead of per second). Read the unit on every choice.

Common indices worth knowing cold

Material	n	Material	n
Ice	1.31	Light flint glass	1.58
Water	1.33	Polycarbonate	1.586
CR-39	1.498	Dense flint	1.92
Crown glass	1.52	Diamond	2.42

> **Trap:** Polycarbonate and light flint both sit near 1.58, so a speed of 117,721 mi/s fits both. When that happens, go with whatever the question pairs it with.

Theories of light, in one line each

- **Corpuscular / Particle (Newton):** light is tiny particles called *corpuscles*. This is also the Emission Theory.
- **Wave / Undulatory (Huygens):** light travels as waves. Young's double-slit later backed this up.
- **Electromagnetic (Maxwell):** light is an EM wave.
- **Quantum (Planck, Einstein):** light comes in energy packets (quanta / photons).

EM spectrum order

`Radio → Micro → IR → Visible → UV → X-ray → Gamma`

Left to right: wavelength shrinks, frequency and energy climb. Radio has the longest wavelength, gamma the shortest. Only the visible band fires the retinal receptors.

> **Hook:** "Raging Martians Invade Venus Using X-ray Guns." Long-and-lazy on the left, short-and-deadly on the right.

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2 · Refraction & Snell's Law

Which way does the ray bend?

- Entering a **denser** medium (higher n): light **slows down and bends toward the normal**.

- Entering a **rarer** medium (lower n): light **speeds up and bends away from the normal**.

> **Hook:** "Slow toward, fast away." The ray always leans toward the slower side, the way a car's wheels pull toward whichever side hits mud first.

Snell's Law

$n_1 \sin \theta_1 = n_2 \sin \theta_2$

When the first medium is air (n=1), this collapses to the handy form `n = sin(incidence) / sin(refraction)`, which is how you find an unknown index from two measured angles.

Apparent depth (the fish and the coin)

`apparent depth = real depth / n` (viewing from air into a denser medium)

An object underwater looks **shallower** than it really is, because the rays bend away from the normal as they leave the water and your eye traces them straight back.

> **Spear-fisher rule:** the fish is always **deeper** than it looks, so aim low. The image is raised; the real thing sits below it.

> **Direction trap:** dividing by n raises the image (object in dense medium seen from air). Multiplying by n is the reverse case (looking from water up into air). Mixing these up is the single most common error in this topic, and the source quiz got the seagull problem wrong this way.

Parallel-sided slab

A ray entering a flat glass slab bends once going in and bends back the same amount coming out, so it leaves **parallel to where it started, just shifted sideways** (lateral displacement). The exit angle equals the entry angle no matter the thickness or index.

`lateral shift d = t · sin(i - r) / cos r`

t is slab thickness, i is incidence, r is the refraction angle inside.

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3 · Lens Power & Focal Length

Power and focal length are reciprocals

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P (diopters) = 1 / f (meters)
```

Keep f in **meters**. A +3.00 D lens has f = 1/3 = 0.33 m. A −8.50 D lens has f = 1/8.50 = 0.12 m.

> **Trap:** if the focal length is given in inches, convert first: `P = 39.37 / f(inches)`. An 8-inch lens is about 4.92 D, not 8 D.

Sign tells you the lens type

- **Plus (+):** converging, convex, thick in the middle.
- **Minus (−):** diverging, concave, thin in the middle.

> **Trap:** if a problem says "diverging" the answer is negative, even when the arithmetic gives a positive number first. Compute the magnitude, then attach the sign.

Lensmaker's equation

$$P = (n - 1)(1/r_1 - 1/r_2)$$

Radii in meters. Each surface alone has power (n−1)/r. Higher index gives more power for the same curve, which is why high-index lenses can be made thinner.

> **Meniscus shortcut:** when the two radii are close in size, their powers nearly cancel and you get a very weak, near-plano lens with a huge focal length. That is why the 45 cm / 55 cm meniscus came out around 435 cm.

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4 · Rectilinear Propagation, Shadows & Pinhole

The master ratio (point source)

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shadow size = object size × (source-to-screen / source-to-object)
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Light from a point travels in straight lines (rectilinear propagation), so the object and its shadow form similar triangles sharing the source as their tip. Every distance is measured **from the source**.

> **Trap:** "Screen is X away from the object" is not the same as "from the source." Add the object distance first. The arrow problem fails if you use 325 instead of 525.

> **Rearrange freely:** the same ratio finds the object size, the screen distance, or the magnification. Magnification = far distance / near distance, both from the source.

Moving the object

- Toward the **screen** → shadow **shrinks**.
- Toward the **source** → shadow **grows**.

Same rule runs the pinhole camera: bring the object closer to the hole and the image gets larger.

Extended source: umbra vs penumbra

- **Umbra** = full dark shadow. **Penumbra** = partial shadow rim.
- Object **same size** as source → umbra stays the **same size** as the object (edges run parallel).
- Object **larger** than source → umbra **widens** with distance.
- Object **smaller** than source → umbra **narrows** to a point, then disappears.
- Spread all three apart → umbra **shrinks**, penumbra grows.

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5 · Photometry

Know the four quantities apart

Quantity	What it is	Unit
Luminous flux	total rate of light energy from a source	lumen
Luminous intensity	flux per solid angle (candle power)	candela
Illuminance	flux landing per unit area on a surface	lux / foot-candle

Solid angle	angular size of a cone, $\omega = A/d^2$	steradian

> **Trap:** if a choice has the wrong unit (steradians or candela for an illuminance question), eliminate it on sight.

Inverse square law

$E = I / d^2$

Illuminance drops with the square of distance. Double the distance and the light falls to a quarter. A 100 cd lamp at 4 m gives 100/16 = 6.25 lux.

Photometer balance (Law of Intensity)

$I_1 / d_1^2 = I_2 / d_2^2$

When two lamps light a screen equally, their candle powers are proportional to the squares of their distances. The **farther** lamp must be the **brighter** one.

> **Sanity check:** if your answer makes the closer lamp brighter, you flipped the ratio.

Solving for distance or height

$d = \sqrt{I / E}$

> **Trap:** do not forget the square root. The tempting wrong answer is always I/E itself (the un-rooted value). The 160-candle lamp sits at 3.51 ft, not 12.31 ft.

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6 · Prisms

Definition of one prism diopter

1Δ = 1 cm of displacement at 1 m → Δ = displacement(cm) / distance(m)

Light bends toward the base; the image shifts toward the apex.

Degrees to prism diopters

`1° ≈ 1.75Δ` (exact: $\Delta = 100 \cdot \tan\theta$)

So 5° is about 8.75Δ, not 10. Use the 1.75 factor for small angles.

Prentice's rule (prism from a lens)

`Δ = F × d` where d = decentration in cm

Rearranged: decentration (cm) = Δ / F. A +5.00 D lens needs 2/5 = 0.4 cm = 4 mm to make 2Δ.

Base direction (the part everyone confuses)

- **Minus lens:** base is in the **same** direction as the viewing point from the OC. Look up → base up. Look temporal → base out.
- **Plus lens:** base is **opposite** the viewing point from the OC. Look up → base down. Look temporal → base in.
- **Decentering the lens** moves the OC, so the line of sight ends up on the opposite side of the OC. Decenter a plus lens up → the eye looks below the OC → base up.

Why: a plus lens bends rays toward its center, a minus lens pushes them away. The base points whichever way the light is being bent.

> **Quick model:** a plus lens is two prisms base-to-base (bases meet at the center); a minus lens is two prisms apex-to-apex (bases out at the edges). Picture which prism your gaze passes through.

> **Trap:** the source quiz keyed "+4.00 D viewed temporal = base out." That is wrong. A plus lens viewed temporal to the OC gives **base in**. The magnitude (4Δ) was right.

Strabismus correcting prism

- **Esotropia** (eye turned in) → correct with **base out**.
- **Exotropia** (eye turned out) → correct with **base in**.

> **Hook:** the base goes opposite the eye's deviation. Eye in, base out. Eye out, base in.

> **Trap:** a cosmetic prism for a blind (non-seeing) eye follows a different goal and can break this rule, which is why the non-seeing exotropia item was keyed base out.

Vocabulary to lock in

- **Apex**: where the two refracting surfaces meet. **Base**: opposite side.
- **Axis of the prism**: the line midway between apex and base, parallel to the apex edge.
- **Centrad / Dennett's method**: a prism that deviates light by 1/100 of a radian.
- **Optical cross**: two perpendicular lines showing the two principal-meridian powers.

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7 · Transposition

Sphero-cylinder to the other sphero-cylinder form

1. New sphere = old sphere + old cylinder
2. New cylinder = same number, **flip the sign**
3. New axis = old axis **± 90°** (keep it between 1 and 180)

> **Hook:** "Add, flip, ninety." Do all three or the answer is wrong.

Crossed cylinder to sphero-cylinder

Pick one cylinder power as the **sphere**. The new **cylinder** is the difference of the two powers. The **axis** is that of the cylinder you turned into the sphere's partner. Example: +1.75 ⊂90 and +1.25 ⊂180 → sphere +1.75, cyl (1.25 − 1.75) = −0.50, axis 180.

Sphero-cylinder to crossed cylinder

Each principal meridian becomes its own cylinder written at the **perpendicular** axis. One meridian carries the sphere alone; the other carries sphere + cylinder. Example: −4.50 sph −2.75 ⊂60 → −4.50 ⊂150 and −7.25 ⊂60.

> **Trap:** the cylinder's power acts **90° away from its written axis**. Always cross the axis when you split the meridians, or the two halves end up on the wrong sides.

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8 · Meridian Power & Multifocals

Power along any meridian

`P at meridian θ = Cyl × sin²(θ - axis)`

A cylinder gives zero power along its own axis and full power 90° away. For −3.50 ⊂180, the 120° meridian is 60° off the axis: $-3.50 \times \sin^2 60° = -3.50 \times 0.75 = -2.62$ D.

> **Two anchors:** at the axis, power = 0. At 90° from the axis, power = full cyl. Everything in between follows the $\sin^2$ curve.

Power in a principal meridian (sphere + cyl)

A cylinder written at axis 90 acts in the horizontal (180) meridian, and vice versa. Horizontal power of −2.00 sph −0.50 ⊂90 = −2.00 + (−0.50) = −2.50 D.

Trifocal intermediate add

`intermediate add = ½ × near add`

Apply that half-add to the distance sphere. A −2.00 distance with a +2.00 near add reads −1.00 at the intermediate.

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9 · Lens Image Formation & Aberrations

Thin-lens and magnification

`1/v - 1/u = 1/f` `m = v / u`

Mind your signs. A concave lens (f negative) always lands a small virtual image on the same side as the object.

What each lens does to the image

- **Concave (minus): always** virtual, erect, minified. No exceptions, no inverted image, ever.
- **Convex (plus):** depends on object distance. A real, inverted image proves the lens is convex.
- **Object exactly at F of a plus lens:** rays leave parallel, image forms at infinity, so effectively **no image**. Watch for the object distance equaling 1/P.

> **Reflex answer:** "concave lens, nature of image?" is virtual-erect-minified every time. Do not compute, just answer.

Image vocabulary

- **Inverted**: upside down. **Perverted**: upright but left-right reversed (a mirror image).
- **Real**: rays actually meet, can be projected. **Virtual**: rays only appear to come from there.

Abbe value and aberrations

Abbe value is the reciprocal of dispersive power. Low Abbe → high dispersion → **high chromatic aberration** (color fringing). Abbe says nothing about spherical aberration, so do not get baited into that choice.

Branches of optics

- **Geometrical optics:** image formation by rays, mirrors, lenses, prisms.
- **Physical optics:** wave behavior (interference, diffraction, polarization).
- **Physiological optics:** the eye and vision itself.

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10 · Reflection & Mirrors

The two laws

- Angle of incidence = angle of reflection, both measured **from the normal** (the perpendicular to the surface).
- The incident ray, reflected ray, and normal all lie in one plane.

Specular reflection needs a smooth surface (mirror). **Diffuse** reflection happens on rough surfaces (paper, wood, concrete), which scatter the light.

Mirror type and focal length

Mirror	Behavior	Focal length
Concave	converging, real focus in front	positive, f = R/2
Convex	diverging, virtual focus behind	negative, f = R/2

> **f = R/2:** the focal point sits halfway to the center of curvature. The center of curvature is the center of the sphere the mirror came from, at distance R.

Concave mirror image, by object position

- Beyond C → real, inverted, diminished.
- At C → real, inverted, **same size**.
- Between C and F → real, inverted, magnified.
- Inside F → **virtual, erect, magnified** (the shaving / make-up mirror).

A convex mirror skips all that: it is **always** virtual, erect, diminished, which is why it is the car side mirror with the wide view.

Three ray-tracing rules

- Parallel ray → reflects through F.
- Ray through F → reflects out parallel.
- Ray to the center of curvature → retraces its own path.

Clinical mirror facts

- **Purkinje images:** the 4th (back of the lens) is the only inverted one.
- **Retinoscopy** reads the reflex off the **retina**; the retinoscope is the instrument.
- **Keratometry** uses the corneal reflection, not the retinal one.

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11 · Critical Angle & Total Internal Reflection

Critical angle

`sin θc = 1 / n` (for a medium-to-air boundary)

The critical angle is the incidence angle in the **denser** medium at which the refracted ray bends to exactly **90°** and grazes the surface. Past it, all the light reflects back inside (total internal reflection).

> **Pattern:** higher index → smaller critical angle. Diamond's tiny critical angle is why it sparkles, since light keeps bouncing inside before it finds a way out.

> **Trap:** "Refraction angle equals what at the critical angle?" The answer is 90°, not 0° and not the same as incidence.
